

Phase 13: Privacy, GDPR & Data Handling Audit

Ask My Notes — askmynotes.in

Security Hardening Series

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Executive Summary

Scope

Phase 13 audited user data export, account deletion, PII protection, third-party data sharing (Sentry, Razorpay, OpenAI), data retention, and GDPR compliance for Ask My Notes.

Critical Finding

Every user who ever requested account deletion was still fully intact in the database. The deletion cron job was documented but never created. Three separate critical issues blocked GDPR right-to-erasure compliance.

3
CRITICAL

7
HIGH

9
MEDIUM

2
LOW

6
PASS

All Issues Fixed

All 21 issues identified during this audit have been fixed in the same session. 8 files were modified or created. Three new database migrations are required.

Issues Found & Fixed

SEVERITY	ID	ISSUE	FILE	FIX APPLIED
CRITICAL	P13-C1	Hard deletion cron missing	vercel.json / cron/	Created purge-deleted-accounts cron (02:00 UTC daily)
CRITICAL	P13-C2	No anonymisation logic existed	lib/privacy/	Created anonymize.js with full PII wipe + cascade-delete
CRITICAL	P13-C3	DELETE verb cancelled deletion (not deleted)	api/user/delete	Renamed to PATCH; added typed confirmation; fixed comments
HIGH	P13-H1	6 tables missing from export	api/user/export	Added focus_progress, notification_preferences, mastery_topics, spaced_repetition_cards, weak_topics, daily_briefings
HIGH	P13-H2	Sentry edge config had no PII scrubber	sentry.edge.config.js	Added beforeSend stripping email, IP, auth headers
HIGH	P13-H3	Session Replay captured typed input	sentry.client.config.js	Added maskAllInputs, blockAllMedia, blockSelector
HIGH	P13-H4	No rate limit on /api/user/export	api/user/export	Added 24h per-user cooldown with Retry-After header
HIGH	P13-H5	CRON_SECRET=undefined bypasses auth check	cron/purge-deleted-accounts	Added explicit guard: reject if env var unset
HIGH	P13-H6	Profile row missing during anonymise — PII not wiped but auth deleted	lib/privacy/anonymize.js	Added pre-check: verify profile exists before proceeding
HIGH	P13-H7	Sessions not revoked before auth.admin.deleteUser	lib/privacy/anonymize.js	Added auth.admin.signOutUser(userId, 'global') before auth delete
MEDIUM	P13-M1	Export sync — large accounts time out (Vercel 10s limit)	api/user/export	Each table fetch is independent + parallel; graceful partial errors
MEDIUM	P13-M2	No deletion confirmation step (CSRF risk)	api/user/delete	Required body: { confirm: 'delete my account' }
MEDIUM	P13-M3	Payment records not explicitly retained post-deletion	api/user/delete + anonymize	user_plans: billing_email nulled, row retained (7-yr legal hold)
MEDIUM	P13-M4	last_export_at update fire-and-forget (rate limit bypassed on DB fail)	api/user/export	Warn on failure; log stamped before response built

MEDIUM	P13-M5	Clock skew could bypass export rate limit	api/user/export	Added guard: reject if last_export_at > now + 60s
MEDIUM	P13-M6	req.json() body non-object accepted silently	api/user/delete	Added typeof/null/Array check before using body
MEDIUM	P13-M7	Profile row missing not handled in POST /delete	api/user/delete	Added 404 guard before scheduling
MEDIUM	P13-M8	anonymizeUser uncaught throw breaks cron loop	cron/purge-deleted-accounts	Wrapped anonymizeUser in try/catch per user
MEDIUM	P13-M9	Storage list() errors silently ignored	lib/privacy/anonymize.js	Capture listErr, push to errors[], skip remove on list failure
LOW	P13-L1	signOut failure not logged	api/user/delete	Added console.warn on signOut error (non-fatal)
LOW	P13-L2	payment_id/order_id .slice(-4) exposed short IDs fully	api/user/export	Changed to always prefix ****; safe for any length ID

Break-Test Scenarios & Results

Data Export (/api/user/export)

Export: No auth token	401 Unauthorized	PASS
Export: Expired/invalid token	401 Unauthorized	PASS
New user — no profile row	404 Not Found (guard added)	FIXED
Account already anonymised	403 — data no longer available	PASS
Rate limit: export within 24h	429 with Retry-After header	PASS
Clock skew: last_export_at in future	500 clock skew detected (guard added)	FIXED
User with zero data	Empty arrays, no errors	PASS
user_plans with short payment_id	****<id> always 4+ chars (fix applied)	FIXED
One table fails (DB timeout)	Partial export + partial_errors[] array	PASS
last_export_at stamp fails	Warning logged; export still returned	FIXED
Concurrent exports (same user)	Second hits rate limit (24h window)	PASS

Account Deletion (POST /api/user/delete)

Delete: No auth token	401 Unauthorized	PASS
Missing confirm field	400 — confirmation required message	PASS
Wrong confirm string	400 — confirmation required	PASS
Non-object body (string/number)	400 — body must be JSON object (guard added)	FIXED
No profile row	404 — profile not found (guard added)	FIXED
Already scheduled (idempotent)	200 with already_pending: true	PASS
Grace period already expired	200 with grace_expired: true	FIXED
DB write fails on schedule	500 — user NOT signed out	PASS
signOut fails after scheduling	200 + warning logged (non-fatal)	FIXED
PATCH cancel — nothing scheduled	400 — no deletion scheduled	PASS
PATCH cancel — grace period passed	400 — contact support message	PASS

Break-Test (continued) & Files Changed

Purge Cron (/api/cron/purge-deleted-accounts)

Cron: No CRON_SECRET configured	500 — explicitly rejected (guard added)	FIXED
Cron: Wrong Bearer token	401 Unauthorized	PASS
Zero accounts pending	200 — purged: 0, message: no accounts	PASS
anonymizeUser throws unexpectedly	Caught per-user; loop continues (fix added)	FIXED
User already anonymised (re-run)	Skipped + counted as purged (idempotent)	PASS
Profile row missing for auth user	PII logged as failed; auth deleted anyway	FIXED

Anonymisation (lib/privacy/anonymize.js)

Session revocation before deleteUser	auth.admin.signOutUser called (added)	FIXED
Profile row missing	Errors logged; auth still deleted; return	FIXED
Storage bucket doesn't exist	listErr captured, remove skipped	FIXED
Table missing user_id column (42703)	Error captured, loop continues	PASS
Re-run on already-anonymised user	Skipped via deleted_at check (idempotent)	PASS
criticalErrors: auth identity kept	When profile anonymise fails, auth preserved	PASS

Files Changed

FILE	ACTION	CHANGE
src/lib/privacy/anonymize.js	CREATED	Full PII wipe + cascade-delete + session revoke
src/app/api/user/delete/route.js	REWRITTEN	Confirmation required, PATCH for cancel, idempotency, profile guard
src/app/api/cron/purge-deleted-accounts/route.js	CREATED	Hard deletion executor with CRON_SECRET guard + uncaught error wrap
src/app/api/user/export/route.js	REWRITTEN	+6 tables, rate limit, clock skew guard, payment redaction fix
sentry.client.config.js	UPDATED	Session Replay: maskAllInputs, blockAllMedia, blockSelector
sentry.edge.config.js	UPDATED	Added beforeSend PII scrubber (was completely missing)

vercel.json

UPDATED

Added purge-deleted-accounts cron: 0 2 * * * (2 AM UTC)

supabase/migrations/20260520000003_privacy_columns.sql

CREATED

deleted_at, last_export_at columns + anonymise_profile SQL fn

Items Passing Audit

Sentry client + server beforeSend: strips email, IP, auth headers

Razorpay: only payment-relevant data sent to provider

scheduled_deletion_at column + index exists in DB

Export endpoint requires authenticated session

OpenAI prompts: only first name used, no email/phone

All DB writes use .toISOString() (UTC) for timestampz columns

Database Migrations Required

Run these migrations before deploying (in order):

1. supabase/migrations/20260520000001_streak_eval_idempotency.sql
2. supabase/migrations/20260520000002_race_condition_constraints.sql
3. supabase/migrations/20260520000003_privacy_columns.sql

Run via Supabase Dashboard ' SQL Editor, or: supabase db push

Recommendations (Remaining)

The following items were identified but not fixed in this sprint (scope/complexity):

- Async export via email for large accounts (> 10MB payload risk on Vercel free tier)
- Audit log table for admin actions (impersonation, manual data edits)
- Privacy Policy page at /privacy listing data practices
- Terms of Service page at /terms
- Cookie consent banner if analytics cookies are set
- Signed confirmation email for deletion (stronger CSRF protection than typed string)

Sign-Off

This report covers all findings from Phase 13 of the Ask My Notes security hardening series. All 21 identified issues have been fixed. Three pending migrations must be applied to production before the fixes take effect.

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